

Package ‘DatabaseConnector’

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Type Package

Title Connecting to Various Database Platforms

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Description An R 'DataBase Interface' ('DBI') compatible interface to various database platforms ('PostgreSQL', 'Oracle', 'Microsoft SQL Server', 'Amazon Redshift', 'Microsoft Parallel Database Warehouse', 'IBM Netezza', 'Apache Impala', 'Google BigQuery', 'Snowflake', 'Spark', 'SQLite', and 'InterSystems IRIS'). Also includes support for fetching data as 'Andromeda' objects. Uses either 'Java Database Connectivity' ('JDBC') or other 'DBI' drivers to connect to databases.

SystemRequirements Java (≥ 8)

Depends R ($\geq 4.0.0$)

Imports rJava,
SqlRender ($\geq 1.19.7$),
methods,
stringr,
readr,
rlang,
utils,
DBI ($\geq 1.0.0$),
urltools,
bit64,
checkmate,
digest,
dbplyr ($\geq 2.6.0$)

Suggests aws.s3,
R.utils,
withr,
testthat,
DBItest,
knitr,
rmarkdown,
RSQLite,
ssh,
Andromeda,
dplyr,
RPostgres,

odbc,
 duckdb,
 bigrquery,
 pool,
 ParallelLogger,
 AzureStor,
 data.table

License Apache License

VignetteBuilder knitr

URL <https://ohdsi.github.io/DatabaseConnector/>, <https://github.com/OHDSI/DatabaseConnector>

BugReports <https://github.com/OHDSI/DatabaseConnector/issues>

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assertTempEmulationSchemaSet

Assert the temp emulation schema is set

Description

Asserts the temp emulation schema is set for DBMSs requiring temp table emulation.

If you know your code uses temp tables, it is a good idea to call this function first, so it can throw an informative error if the user forgot to set the temp emulation schema.

Usage

```
assertTempEmulationSchemaSet(
  dbms,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema")
)
```

Arguments

dbms The type of DBMS running on the server. See [connect\(\)](#) or [createConnectionDetails\(\)](#) for valid values.

tempEmulationSchema The temp emulation schema specified by the user.

Value

Does not return anything. Throws an error if the DBMS requires temp emulation but the temp emulation schema is not set.

computeDataHash	<i>Compute hash of data</i>
-----------------	-----------------------------

Description

Compute a hash of the data in the database schema. If the data changes, this should produce a different hash code. Specifically, the hash is based on the field names, field types, and table row counts.

Usage

```
computeDataHash(connection, databaseSchema, tables = NULL, progressBar = TRUE)
```

Arguments

connection The connection to the database server created using either [connect\(\)](#) or [DBI::dbConnect\(\)](#).

databaseSchema The name of the database schema. See details for platform-specific details.

tables (Optional) A list of tables to restrict to.

progressBar When true, a progress bar is shown based on the number of tables in the database schema.

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

A string representing the MD5 hash code.

connect	<i>connect</i>
---------	----------------

Description

Creates a connection to a database server .There are four ways to call this function:

- `connect(dbms, user, password, server, port, extraSettings, oracleDriver, pathToDriver)`
- `connect(connectionDetails)`
- `connect(dbms, connectionString, pathToDriver))`
- `connect(dbms, connectionString, user, password, pathToDriver)`

DBMS parameter details::

Depending on the DBMS, the function arguments have slightly different interpretations:

Oracle:

- `user`. The user name used to access the server
- `password`. The password for that user
- `server`. This field contains the SID, or host and servicename, SID, or TNSName: 'sid', 'host/sid', 'host/service name', or 'tnsname'
- `port`. Specifies the port on the server (default = 1521)
- `extraSettings`. The configuration settings for the connection (i.e. SSL Settings such as "(PROTOCOL=tcp)")
- `oracleDriver`. The driver to be used. Choose between "thin" or "oci".
- `pathToDriver`. The path to the folder containing the Oracle JDBC driver JAR files.

Microsoft SQL Server:

- `user`. The user used to log in to the server. If the user is not specified, Windows Integrated Security will be used, which requires the SQL Server JDBC drivers to be installed (see details below).
- `password`. The password used to log on to the server
- `server`. This field contains the host name of the server
- `port`. Not used for SQL Server
- `extraSettings`. The configuration settings for the connection (i.e. SSL Settings such as "encrypt=true; trustServerCertificate=false;")
- `pathToDriver`. The path to the folder containing the SQL Server JDBC driver JAR files.

Microsoft PDW:

- `user`. The user used to log in to the server. If the user is not specified, Windows Integrated Security will be used, which requires the SQL Server JDBC drivers to be installed (see details below).
- `password`. The password used to log on to the server
- `server`. This field contains the host name of the server
- `port`. Not used for SQL Server
- `extraSettings`. The configuration settings for the connection (i.e. SSL Settings such as "encrypt=true; trustServerCertificate=false;")
- `pathToDriver`. The path to the folder containing the SQL Server JDBC driver JAR files.

PostgreSQL:

- `user`. The user used to log in to the server
- `password`. The password used to log on to the server
- `server`. This field contains the host name of the server and the database holding the relevant schemas: `host/database`
- `port`. Specifies the port on the server (default = 5432)
- `extraSettings`. The configuration settings for the connection (i.e. SSL Settings such as `"ssl=true"`)
- `pathToDriver`. The path to the folder containing the PostgreSQL JDBC driver JAR files.

Redshift:

- `user`. The user used to log in to the server
- `password`. The password used to log on to the server
- `server`. This field contains the host name of the server and the database holding the relevant schemas: `host/database`
- `port`. Specifies the port on the server (default = 5439)
- `extraSettings`. The configuration settings for the connection (i.e. SSL Settings such as `"ssl=true&sslfactory=com.amazon.redshift.jdbc.Driver"`)
- `pathToDriver`. The path to the folder containing the RedShift JDBC driver JAR files.

Netezza:

- `user`. The user used to log in to the server
- `password`. The password used to log on to the server
- `server`. This field contains the host name of the server and the database holding the relevant schemas: `host/database`
- `port`. Specifies the port on the server (default = 5480)
- `extraSettings`. The configuration settings for the connection (i.e. SSL Settings such as `"ssl=true"`)
- `pathToDriver`. The path to the folder containing the Netezza JDBC driver JAR file (`nzjdbc.jar`).

Impala:

- `user`. The user name used to access the server
- `password`. The password for that user
- `server`. The host name of the server
- `port`. Specifies the port on the server (default = 21050)
- `extraSettings`. The configuration settings for the connection (i.e. SSL Settings such as `"SSLKeyStorePwd=*****"`)
- `pathToDriver`. The path to the folder containing the Impala JDBC driver JAR files.

SQLite:

- `server`. The path to the SQLite file.

DuckDB:

- `server`. The path to the DuckDB file.
- `extraSettings`. Additional settings for DuckDB. For DuckDB specifically, if `extraSettings$config` is provided, it will be passed to the `duckdb::duckdb()` constructor. For example: `extraSettings = list(config = list(memory_limit = "8GB", preserve_insertion_order = "false"))`

Spark / Databricks:

Currently both JDBC and ODBC connections are supported for Spark. Set the `connectionString` argument to use JDBC, otherwise ODBC is used:

- `connectionString`. The JDBC connection string (e.g. something like `'jdbc:databricks://my-org.cloud.databricks.com:443/default;transportMode=http;ssl=1;AuthMech=3;httpPath=/sql/1.0/warehouses/abc'`)

- `user`. The user name used to access the server. This can be set to 'token' when using a personal token (recommended).
- `password`. The password for that user. This should be your personal token when using a personal token (recommended).
- `server`. The host name of the server (when using ODBC), e.g. 'my-org.cloud.databricks.com')
- `port`. Specifies the port on the server (when using ODBC)
- `extraSettings`. Additional settings for the ODBC connection, for example `extraSettings = list(HTTPPath = "/sql/1.0/warehouses/abcde12345", SSL = 1, ThriftTransport = 2, AuthMech = 3)`

Snowflake:

- `connectionString`. The connection string (e.g. starting with 'jdbc:snowflake://host:port/?db=database').
- `user`. The user name used to access the server.
- `password`. The password for that user.

InterSystems IRIS:

- `connectionString`. The connection string (e.g. starting with 'jdbc:IRIS://host:port/namespace'). Alternatively, you can provide values for server and port, in which case the default USER namespace is used to connect.
- `user`. The user name used to access the server.
- `password`. The password for that user.
- `pathToDriver`. The path to the folder containing the InterSystems IRIS JDBC driver JAR file.

Windows authentication for SQL Server::

To be able to use Windows authentication for SQL Server (and PDW), you have to install the JDBC driver. Download the version 9.2.0 .zip from [Microsoft](#) and extract its contents to a folder. In the extracted folder you will find the file `sqljdbc_9.2/enu/auth/x64/mssql-jdbc_auth-9.2.0.x64.dll` (64-bits) or `sqljdbc_9.2/enu/auth/x86/mssql-jdbc_auth-9.2.0.x86.dll` (32-bits), which needs to be moved to location on the system path, for example to `c:/windows/system32`. If you not have write access to any folder in the system path, you can also specify the path to the folder containing the dll by setting the environmental variable `PATH_TO_AUTH_DLL`, so for example `Sys.setenv("PATH_TO_AUTH_DLL" = "c:/temp")` Note that the environmental variable needs to be set before calling `connect()` for the first time.

Arguments

`connectionDetails`

An object of class `connectionDetails` as created by the `createConnectionDetails()` function.

`dbms`

The type of DBMS running on the server. Valid values are

- "oracle" for Oracle
- "postgresql" for PostgreSQL
- "redshift" for Amazon Redshift
- "sql server" for Microsoft SQL Server
- "pdw" for Microsoft Parallel Data Warehouse (PDW)
- "netezza" for IBM Netezza
- "bigquery" for Google BigQuery
- "sqlite" for SQLite
- "sqlite extended" for SQLite with extended types (DATE and DATETIME)

	• "duckdb" for DuckDB • "spark" for Spark • "snowflake" for Snowflake • "iris" for InterSystems IRIS
user	The user name used to access the server.
password	The password for that user.
server	The name of the server.
port	(optional) The port on the server to connect to.
extraSettings	(optional) Additional configuration settings specific to the database provider to configure things as security for SSL. For connections using JDBC these will be appended to end of the connection string. For connections using DBI, these settings will additionally be used to call DBI::dbConnect() .
oracleDriver	Specify which Oracle drive you want to use. Choose between "thin" or "oci".
connectionString	The JDBC connection string. If specified, the server, port, extraSettings, and oracleDriver fields are ignored. If user and password are not specified, they are assumed to already be included in the connection string.
pathToDriver	Path to a folder containing the JDBC driver JAR files. See downloadJdbcDrivers() for instructions on how to download the relevant drivers.

Details

This function creates a connection to a database.

Value

An object that extends DBIConnection in a database-specific manner. This object is used to direct commands to the database engine.

Examples

```
## Not run:
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = "localhost/postgres",
  user = "root",
  password = "xxx"
)
conn <- connect(connectionDetails)
dbGetQuery(conn, "SELECT COUNT(*) FROM person")
disconnect(conn)

conn <- connect(dbms = "sql server", server = "RNDUSRDHIT06.jnj.com")
dbGetQuery(conn, "SELECT COUNT(*) FROM concept")
disconnect(conn)

conn <- connect(
  dbms = "oracle",
  server = "127.0.0.1/xe",
  user = "system",
  password = "xxx",
  pathToDriver = "c:/temp"
```

```

)
dbGetQuery(conn, "SELECT COUNT(*) FROM test_table")
disconnect(conn)

conn <- connect(
  dbms = "postgresql",
  connectionString = "jdbc:postgresql://127.0.0.1:5432/cmd_database"
)
dbGetQuery(conn, "SELECT COUNT(*) FROM person")
disconnect(conn)

## End(Not run)

```

```
createConnectionDetails
```

```
createConnectionDetails
```

Description

Creates a list containing all details needed to connect to a database. There are three ways to call this function:

- `createConnectionDetails(dbms, user, password, server, port, extraSettings, oracleDriver, pathToDriver)`
- `createConnectionDetails(dbms, connectionString, pathToDriver)`
- `createConnectionDetails(dbms, connectionString, user, password, pathToDriver)`

DBMS parameter details::

Depending on the DBMS, the function arguments have slightly different interpretations:

Oracle:

- `user`. The user name used to access the server
- `password`. The password for that user
- `server`. This field contains the SID, or host and servicename, SID, or TNSName: 'sid', 'host/sid', 'host/service name', or 'tnsname'
- `port`. Specifies the port on the server (default = 1521)
- `extraSettings`. The configuration settings for the connection (i.e. SSL Settings such as "(PROTOCOL=tcps)")
- `oracleDriver`. The driver to be used. Choose between "thin" or "oci".
- `pathToDriver`. The path to the folder containing the Oracle JDBC driver JAR files.

Microsoft SQL Server:

- `user`. The user used to log in to the server. If the user is not specified, Windows Integrated Security will be used, which requires the SQL Server JDBC drivers to be installed (see details below).
- `password`. The password used to log on to the server
- `server`. This field contains the host name of the server
- `port`. Not used for SQL Server
- `extraSettings`. The configuration settings for the connection (i.e. SSL Settings such as "encrypt=true; trustServerCertificate=false;")
- `pathToDriver`. The path to the folder containing the SQL Server JDBC driver JAR files.

Microsoft PDW:

- **user.** The user used to log in to the server. If the user is not specified, Windows Integrated Security will be used, which requires the SQL Server JDBC drivers to be installed (see details below).
- **password.** The password used to log on to the server
- **server.** This field contains the host name of the server
- **port.** Not used for SQL Server
- **extraSettings.** The configuration settings for the connection (i.e. SSL Settings such as "encrypt=true; trustServerCertificate=false;")
- **pathToDriver.** The path to the folder containing the SQL Server JDBC driver JAR files.

PostgreSQL:

- **user.** The user used to log in to the server
- **password.** The password used to log on to the server
- **server.** This field contains the host name of the server and the database holding the relevant schemas: host/database
- **port.** Specifies the port on the server (default = 5432)
- **extraSettings.** The configuration settings for the connection (i.e. SSL Settings such as "ssl=true")
- **pathToDriver.** The path to the folder containing the PostgreSQL JDBC driver JAR files.

Redshift:

- **user.** The user used to log in to the server
- **password.** The password used to log on to the server
- **server.** This field contains the host name of the server and the database holding the relevant schemas: host/database
- **port.** Specifies the port on the server (default = 5439)
- **extraSettings** The configuration settings for the connection (i.e. SSL Settings such as "ssl=true&sslfactory=com.amazonaws.redshift.jdbc42.SSLConnectionFactory")
- **pathToDriver.** The path to the folder containing the RedShift JDBC driver JAR files.

Netezza:

- **user.** The user used to log in to the server
- **password.** The password used to log on to the server
- **server.** This field contains the host name of the server and the database holding the relevant schemas: host/database
- **port.** Specifies the port on the server (default = 5480)
- **extraSettings.** The configuration settings for the connection (i.e. SSL Settings such as "ssl=true")
- **pathToDriver.** The path to the folder containing the Netezza JDBC driver JAR file (nzjdbc.jar).

Impala:

- **user.** The user name used to access the server
- **password.** The password for that user
- **server.** The host name of the server
- **port.** Specifies the port on the server (default = 21050)
- **extraSettings.** The configuration settings for the connection (i.e. SSL Settings such as "SSLKeyStorePwd=*****")
- **pathToDriver.** The path to the folder containing the Impala JDBC driver JAR files.

SQLite:

- `server`. The path to the SQLite file.

DuckDB:

- `server`. The path to the DuckDB file.
- `extraSettings`. Additional settings for DuckDB. For DuckDB specifically, if `extraSettings$config` is provided, it will be passed to the `duckdb::duckdb()` constructor. For example: `extraSettings = list(config = list(memory_limit = "8GB", preserve_insertion_order = "false"))`

Spark / Databricks:

Currently both JDBC and ODBC connections are supported for Spark. Set the `connectionString` argument to use JDBC, otherwise ODBC is used:

- `connectionString`. The JDBC connection string (e.g. something like `'jdbc:databricks://my-org.cloud.databricks.com:443/default;transportMode=http;ssl=1;AuthMech=3;httpPath=/sql/1.0/warehouses/abcde12345'`).
- `user`. The user name used to access the server. This can be set to `'token'` when using a personal token (recommended).
- `password`. The password for that user. This should be your personal token when using a personal token (recommended).
- `server`. The host name of the server (when using ODBC), e.g. `'my-org.cloud.databricks.com'`
- `port`. Specifies the port on the server (when using ODBC)
- `extraSettings`. Additional settings for the ODBC connection, for example `extraSettings = list(HTTPPath = "/sql/1.0/warehouses/abcde12345", SSL = 1, ThriftTransport = 2, AuthMech = 3)`

Snowflake:

- `connectionString`. The connection string (e.g. starting with `'jdbc:snowflake://host:port/?db=database'`).
- `user`. The user name used to access the server.
- `password`. The password for that user.

InterSystems IRIS:

- `connectionString`. The connection string (e.g. starting with `'jdbc:IRIS://host:port/namespace'`). Alternatively, you can provide values for `server` and `port`, in which case the default `USER` namespace is used to connect.
- `user`. The user name used to access the server.
- `password`. The password for that user.
- `pathToDriver`. The path to the folder containing the InterSystems IRIS JDBC driver JAR file.

Windows authentication for SQL Server::

To be able to use Windows authentication for SQL Server (and PDW), you have to install the JDBC driver. Download the version 9.2.0 .zip from [Microsoft](#) and extract its contents to a folder. In the extracted folder you will find the file `sqljdbc_9.2/enu/auth/x64/mssql-jdbc_auth-9.2.0.x64.dll` (64-bits) or `sqljdbc_9.2/enu/auth/x86/mssql-jdbc_auth-9.2.0.x86.dll` (32-bits), which needs to be moved to location on the system path, for example to `c:/windows/system32`. If you not have write access to any folder in the system path, you can also specify the path to the folder containing the dll by setting the environmental variable `PATH_TO_AUTH_DLL`, so for example `Sys.setenv("PATH_TO_AUTH_DLL" = "c:/temp")` Note that the environmental variable needs to be set before calling `connect()` for the first time.

Arguments

- | | |
|-------------------|--|
| <code>dbms</code> | The type of DBMS running on the server. Valid values are <ul style="list-style-type: none"> • "oracle" for Oracle |
|-------------------|--|

	• "postgresql" for PostgreSQL • "redshift" for Amazon Redshift • "sql server" for Microsoft SQL Server • "pdw" for Microsoft Parallel Data Warehouse (PDW) • "netezza" for IBM Netezza • "bigquery" for Google BigQuery • "sqlite" for SQLite • "sqlite extended" for SQLite with extended types (DATE and DATETIME) • "duckdb" for DuckDB • "spark" for Spark • "snowflake" for Snowflake • "iris" for InterSystems IRIS
user	The user name used to access the server.
password	The password for that user.
server	The name of the server.
port	(optional) The port on the server to connect to.
extraSettings	(optional) Additional configuration settings specific to the database provider to configure things as security for SSL. For connections using JDBC these will be appended to end of the connection string. For connections using DBI, these settings will additionally be used to call <code>DBI::dbConnect()</code> .
oracleDriver	Specify which Oracle drive you want to use. Choose between "thin" or "oci".
connectionString	The JDBC connection string. If specified, the server, port, extraSettings, and oracleDriver fields are ignored. If user and password are not specified, they are assumed to already be included in the connection string.
pathToDriver	Path to a folder containing the JDBC driver JAR files. See <code>downloadJdbcDrivers()</code> for instructions on how to download the relevant drivers.

Details

This function creates a list containing all details needed to connect to a database. The list can then be used in the `connect()` function.

It is highly recommended to use a secure approach to storing credentials, so not to have your credentials in plain text in your R scripts. The examples demonstrate how to use the keyring package.

Value

A list with all the details needed to connect to a database.

Examples

```
## Not run:
# Needs to be done only once on a machine. Credentials will then be stored in
# the operating system's secure credential manager:
keyring::key_set_with_value("server", password = "localhost/postgres")
keyring::key_set_with_value("user", password = "root")
keyring::key_set_with_value("password", password = "secret")

# Create connection details using keyring. Note: the connection details will
```

```
# not store the credentials themselves, but the reference to get the credentials.
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = keyring::key_get("server"),
  user = keyring::key_get("user"),
  password = keyring::key_get("password"),
)
conn <- connect(connectionDetails)
dbGetQuery(conn, "SELECT COUNT(*) FROM person")
disconnect(conn)

## End(Not run)
```

```
createDbiConnectionDetails
```

```
Create DBI connection details
```

Description

For advanced users only. This function will allow DatabaseConnector to wrap any DBI driver. Using a driver that DatabaseConnector hasn't been tested with may give unpredictable performance. Use at your own risk. No support will be provided.

Usage

```
createDbiConnectionDetails(dbms, drv, ...)
```

Arguments

dbms	The type of DBMS running on the server. Valid values are • "oracle" for Oracle • "postgresql" for PostgreSQL • "redshift" for Amazon Redshift • "sql server" for Microsoft SQL Server • "pdw" for Microsoft Parallel Data Warehouse (PDW) • "netezza" for IBM Netezza • "bigquery" for Google BigQuery • "sqlite" for SQLite • "sqlite extended" for SQLite with extended types (DATE and DATETIME) • "duckdb" for DuckDB • "spark" for Spark • "snowflake" for Snowflake • "iris" for InterSystems IRIS
drv	An object that inherits from DBIDriver, or an existing DBIConnection object (in order to clone an existing connection).
...	authentication arguments needed by the DBMS instance; these typically include user, password, host, port, dbname, etc. For details see the appropriate DBIDriver

Value

A list with all the details needed to connect to a database.

createZipFile	<i>Compress files and/or folders into a single zip file</i>
---------------	---

Description

Compress files and/or folders into a single zip file

Usage

```
createZipFile(zipFile, files, rootFolder = getwd(), compressionLevel = 9)
```

Arguments

zipFile	The path to the zip file to be created.
files	The files and/or folders to be included in the zip file. Folders will be included recursively.
rootFolder	The root folder. All files will be stored with relative paths relative to this folder.
compressionLevel	A number between 1 and 9. 9 compresses best, but it also takes the longest.

Details

Uses Java's compression library to create a zip file. It is similar to `utils::zip`, except that it does not require an external zip tool to be available on the system path.

DatabaseConnectorDriver	<i>Create a DatabaseConnectorDriver object</i>
-------------------------	--

Description

Create a DatabaseConnectorDriver object

Usage

```
DatabaseConnectorDriver()
```

dbAppendTable, DatabaseConnectorConnection, character-method
Insert rows into a table

Description

The dbAppendTable() method assumes that the table has been created beforehand, e.g. with dbCreateTable(). The default implementation calls sqlAppendTableTemplate() and then dbExecute() with the param argument. Use dbAppendTableArrow() to append data from an Arrow stream.

Usage

```
## S4 method for signature 'DatabaseConnectorConnection,character'
dbAppendTable(
  conn,
  name,
  value,
  databaseSchema = NULL,
  temporary = FALSE,
  ...,
  row.names = NULL
)
```

Arguments

conn	A DBI::DBConnection object, as returned by dbConnect() .
name	The table name, passed on to dbQuoteIdentifier() . Options are: - a character string with the unquoted DBMS table name, e.g. "table_name", - a call to Id() with components to the fully qualified table name, e.g. Id(schema = "my_schema", table = "table_name") - a call to SQL() with the quoted and fully qualified table name given verbatim, e.g. SQL('"my_schema"."table_name"')
value	A data.frame (or coercible to data.frame).
databaseSchema	The name of the database schema. See details for platform-specific details.
temporary	Should the table created as a temp table?
...	Other parameters passed on to methods.
row.names	Must be NULL.

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

dbAppendTable() returns a scalar numeric.

See Also

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

dbClearResult, DatabaseConnectorDbiResult-method

Clear a result set

Description

Frees all resources (local and remote) associated with a result set. This step is mandatory for all objects obtained by calling [dbSendQuery\(\)](#) or [dbSendStatement\(\)](#).

Usage

```
## S4 method for signature 'DatabaseConnectorDbiResult'
dbClearResult(res, ...)
```

Arguments

res	An object inheriting from DBI::DBIResult .
...	Other arguments passed on to methods.

Value

[dbClearResult\(\)](#) returns TRUE, invisibly, for result sets obtained from [dbSendQuery\(\)](#), [dbSendStatement\(\)](#), or [dbSendQueryArrow\(\)](#),

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other DBIResultArrow generics: [DBIResultArrow-class](#), [dbBind\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbHasCompleted\(\)](#), [dbIsValid\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbFetch\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbHasCompleted\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#)

Other command execution generics: [dbBind\(\)](#), [dbExecute\(\)](#), [dbGetRowsAffected\(\)](#), [dbSendStatement\(\)](#)

dbClearResult, DatabaseConnectorJdbcResult-method
Clear a result set

Description

Frees all resources (local and remote) associated with a result set. This step is mandatory for all objects obtained by calling [dbSendQuery\(\)](#) or [dbSendStatement\(\)](#).

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcResult'
dbClearResult(res, ...)
```

Arguments

res	An object inheriting from DBI::DBIResult .
...	Other arguments passed on to methods.

Value

dbClearResult() returns TRUE, invisibly, for result sets obtained from dbSendQuery(), dbSendStatement(), or dbSendQueryArrow(),

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other DBIResultArrow generics: [DBIResultArrow-class](#), [dbBind\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbHasCompleted\(\)](#), [dbIsValid\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbFetch\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbHasCompleted\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#)

Other command execution generics: [dbBind\(\)](#), [dbExecute\(\)](#), [dbGetRowsAffected\(\)](#), [dbSendStatement\(\)](#)

dbColumnInfo, DatabaseConnectorDbiResult-method
Information about result types

Description

Produces a data.frame that describes the output of a query. The data.frame should have as many rows as there are output fields in the result set, and each column in the data.frame describes an aspect of the result set field (field name, type, etc.)

Usage

```
## S4 method for signature 'DatabaseConnectorDbiResult'
dbColumnInfo(res, ...)
```

Arguments

res An object inheriting from [DBI::DBIResult](#).
 ... Other arguments passed on to methods.

Value

dbColumnInfo() returns a data frame with at least two columns "name" and "type" (in that order) (and optional columns that start with a dot). The "name" and "type" columns contain the names and types of the R columns of the data frame that is returned from [DBI::dbFetch\(\)](#). The "type" column is of type character and only for information. Do not compute on the "type" column, instead use dbFetch(res, n = 0) to create a zero-row data frame initialized with the correct data types.

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

dbColumnInfo, DatabaseConnectorJdbcResult-method

Information about result types

Description

Produces a data.frame that describes the output of a query. The data.frame should have as many rows as there are output fields in the result set, and each column in the data.frame describes an aspect of the result set field (field name, type, etc.)

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcResult'
dbColumnInfo(res, ...)
```

Arguments

res An object inheriting from [DBI::DBIResult](#).
 ... Other arguments passed on to methods.

Value

dbColumnInfo() returns a data frame with at least two columns "name" and "type" (in that order) (and optional columns that start with a dot). The "name" and "type" columns contain the names and types of the R columns of the data frame that is returned from [DBI::dbFetch\(\)](#). The "type" column is of type character and only for information. Do not compute on the "type" column, instead use dbFetch(res, n = 0) to create a zero-row data frame initialized with the correct data types.

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

dbConnect, DatabaseConnectorDriver-method

Create a connection to a DBMS

Description

Connect to a database. This function is synonymous with the [connect\(\)](#) function. except a dummy driver needs to be specified

Usage

```
## S4 method for signature 'DatabaseConnectorDriver'
dbConnect(drv, ...)
```

Arguments

drv	The result of the DatabaseConnectorDriver() function
...	Other parameters. These are the same as expected by the connect() function.

Value

Returns a DatabaseConnectorConnection object that can be used with most of the other functions in this package.

Examples

```
## Not run:
conn <- dbConnect(DatabaseConnectorDriver(),
  dbms = "postgresql",
  server = "localhost/ohdsi",
  user = "joe",
  password = "secret"
)
querySql(conn, "SELECT * FROM cdm_synpuf.person;")
dbDisconnect(conn)

## End(Not run)
```

dbCreateTable, DatabaseConnectorConnection-method

Create a table in the database

Description

The default `dbCreateTable()` method calls `sqlCreateTable()` and `dbExecute()`. Use `dbCreateTableArrow()` to create a table from an Arrow schema.

Usage

```
## S4 method for signature 'DatabaseConnectorConnection'
dbCreateTable(
  conn,
  name,
  fields,
  databaseSchema = NULL,
  ...,
  row.names = NULL,
  temporary = FALSE
)
```

Arguments

<code>conn</code>	A DBI::DBIConnection object, as returned by dbConnect() .
<code>name</code>	The table name, passed on to dbQuoteIdentifier() . Options are: - a character string with the unquoted DBMS table name, e.g. <code>"table_name"</code>, - a call to Id() with components to the fully qualified table name, e.g. <code>Id(schema = "my_schema", table = "table_name")</code> - a call to SQL() with the quoted and fully qualified table name given verbatim, e.g. <code>SQL('"my_schema"."table_name"')</code>
<code>fields</code>	Either a character vector or a data frame. A named character vector: Names are column names, values are types. Names are escaped with dbQuoteIdentifier() . Field types are unescaped. A data frame: field types are generated using dbDataType() .
<code>databaseSchema</code>	The name of the database schema. See details for platform-specific details.
<code>...</code>	Other parameters passed on to methods.
<code>row.names</code>	Must be <code>NULL</code> .
<code>temporary</code>	Should the table created as a temp table?

Details

The `databaseSchema` argument is interpreted differently according to the different platforms: SQL Server and PDW: The `databaseSchema` schema should specify both the database and the schema, e.g. `'my_database.dbo'`. Impala: the `databaseSchema` should specify the database. Oracle: The `databaseSchema` should specify the Oracle 'user'. All other : The `databaseSchema` should specify the schema.

Value

dbCreateTable() returns TRUE, invisibly.

See Also

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

dbDisconnect, DatabaseConnectorConnection-method

Disconnect (close) a connection

Description

This closes the connection, discards all pending work, and frees resources (e.g., memory, sockets).

Usage

```
## S4 method for signature 'DatabaseConnectorConnection'
dbDisconnect(conn)
```

Arguments

conn A [DBI::DBIConnection](#) object, as returned by [dbConnect\(\)](#).

Value

dbDisconnect() returns TRUE, invisibly.

See Also

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

dbExecute, DatabaseConnectorDbiConnection, character-method

Change database state

Description

Executes a statement and returns the number of rows affected. `dbExecute()` comes with a default implementation (which should work with most backends) that calls `dbSendStatement()`, then `dbGetRowsAffected()`, ensuring that the result is always freed by `dbClearResult()`. For passing query parameters, see `dbBind()`, in particular the "The command execution flow" section.

Usage

```
## S4 method for signature 'DatabaseConnectorDbiConnection,character'
dbExecute(conn, statement, ...)
```

Arguments

<code>conn</code>	A DBI::DBIConnection object, as returned by <code>dbConnect()</code> .
<code>statement</code>	a character string containing SQL.
<code>...</code>	Other parameters passed on to methods.

Details

You can also use `dbExecute()` to call a stored procedure that performs data manipulation or other actions that do not return a result set. To execute a stored procedure that returns a result set, or a data manipulation query that also returns a result set such as `INSERT INTO ... RETURNING ...`, use `dbGetQuery()` instead.

Value

`dbExecute()` always returns a scalar numeric that specifies the number of rows affected by the statement.

See Also

For queries: [dbSendQuery\(\)](#) and [dbGetQuery\(\)](#).

Other `DBIConnection` generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other command execution generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbGetRowsAffected\(\)](#), [dbSendStatement\(\)](#)

dbExecute, DatabaseConnectorJdbcConnection, character-method
Change database state

Description

Executes a statement and returns the number of rows affected. `dbExecute()` comes with a default implementation (which should work with most backends) that calls `dbSendStatement()`, then `dbGetRowsAffected()`, ensuring that the result is always freed by `dbClearResult()`. For passing query parameters, see `dbBind()`, in particular the "The command execution flow" section.

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcConnection,character'
dbExecute(conn, statement, ...)
```

Arguments

<code>conn</code>	A DBI::DBIConnection object, as returned by <code>dbConnect()</code> .
<code>statement</code>	a character string containing SQL.
<code>...</code>	Other parameters passed on to methods.

Details

You can also use `dbExecute()` to call a stored procedure that performs data manipulation or other actions that do not return a result set. To execute a stored procedure that returns a result set, or a data manipulation query that also returns a result set such as `INSERT INTO ... RETURNING ...`, use `dbGetQuery()` instead.

Value

`dbExecute()` always returns a scalar numeric that specifies the number of rows affected by the statement.

See Also

For queries: [dbSendQuery\(\)](#) and [dbGetQuery\(\)](#).

Other `DBIConnection` generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other command execution generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbGetRowsAffected\(\)](#), [dbSendStatement\(\)](#)

dbExistsTable, DatabaseConnectorConnection, character-method

Does a table exist?

Description

Returns if a table given by name exists in the database.

Usage

```
## S4 method for signature 'DatabaseConnectorConnection,character'
dbExistsTable(conn, name, databaseSchema = NULL, ...)
```

Arguments

conn	A DBI::DBIConnection object, as returned by dbConnect() .
name	The table name, passed on to dbQuoteIdentifier() . Options are: • a character string with the unquoted DBMS table name, e.g. "table_name", • a call to Id() with components to the fully qualified table name, e.g. Id(schema = "my_schema", table = "table_name") • a call to SQL() with the quoted and fully qualified table name given verbatim, e.g. SQL('"my_schema"."table_name"')
databaseSchema	The name of the database schema. See details for platform-specific details.
...	Other parameters passed on to methods.

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

dbExistsTable() returns a logical scalar, TRUE if the table or view specified by the name argument exists, FALSE otherwise.

This includes temporary tables if supported by the database.

See Also

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

dbFetch, DatabaseConnectorDbiResult-method

Fetch records from a previously executed query

Description

Fetch the next *n* elements (rows) from the result set and return them as a `data.frame`.

Usage

```
## S4 method for signature 'DatabaseConnectorDbiResult'
dbFetch(res, n = -1, ...)
```

Arguments

<code>res</code>	An object inheriting from DBI::DBIResult , created by dbSendQuery() .
<code>n</code>	maximum number of records to retrieve per fetch. Use <code>n = -1</code> or <code>n = Inf</code> to retrieve all pending records. Some implementations may recognize other special values.
<code>...</code>	Other arguments passed on to methods.

Details

`fetch()` is provided for compatibility with older DBI clients - for all new code you are strongly encouraged to use `dbFetch()`. The default implementation for `dbFetch()` calls `fetch()` so that it is compatible with existing code. Modern backends should implement for `dbFetch()` only.

Value

`dbFetch()` always returns a [data.frame](#) with as many rows as records were fetched and as many columns as fields in the result set, even if the result is a single value or has one or zero rows. Passing `n = NA` is supported and returns an arbitrary number of rows (at least one) as specified by the driver, but at most the remaining rows in the result set.

See Also

Close the result set with [dbClearResult\(\)](#) as soon as you finish retrieving the records you want.

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbHasCompleted\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#)

dbFetch, DatabaseConnectorJdbcResult-method

Fetch records from a previously executed query

Description

Fetch the next *n* elements (rows) from the result set and return them as a data.frame.

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcResult'
dbFetch(res, n = -1, ...)
```

Arguments

<i>res</i>	An object inheriting from DBI::DBIResult , created by dbSendQuery() .
<i>n</i>	maximum number of records to retrieve per fetch. Use <i>n</i> = -1 or <i>n</i> = Inf to retrieve all pending records. Some implementations may recognize other special values.
<i>...</i>	Other arguments passed on to methods.

Details

[fetch\(\)](#) is provided for compatibility with older DBI clients - for all new code you are strongly encouraged to use [dbFetch\(\)](#). The default implementation for [dbFetch\(\)](#) calls [fetch\(\)](#) so that it is compatible with existing code. Modern backends should implement for [dbFetch\(\)](#) only.

Value

[dbFetch\(\)](#) always returns a [data.frame](#) with as many rows as records were fetched and as many columns as fields in the result set, even if the result is a single value or has one or zero rows. Passing *n* = NA is supported and returns an arbitrary number of rows (at least one) as specified by the driver, but at most the remaining rows in the result set.

See Also

Close the result set with [dbClearResult\(\)](#) as soon as you finish retrieving the records you want.

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbHasCompleted\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#)

dbGetInfo, DatabaseConnectorConnection-method
Get DBMS metadata

Description

Retrieves information on objects of class [DBIDriver](#), [DBIConnection](#) or [DBIResult](#).

Usage

```
## S4 method for signature 'DatabaseConnectorConnection'
dbGetInfo(dbObj, ...)
```

Arguments

dbObj	An object inheriting from DBIObject , i.e. DBIDriver , DBIConnection , or a DBIResult
...	Other arguments to methods.

Value

For objects of class [DBI::DBIDriver](#), `dbGetInfo()` returns a named list that contains at least the following components:

- `driver.version`: the package version of the DBI backend,
- `client.version`: the version of the DBMS client library.

For objects of class [DBI::DBIConnection](#), `dbGetInfo()` returns a named list that contains at least the following components:

- `db.version`: version of the database server,
- `dbname`: database name,
- `username`: username to connect to the database,
- `host`: hostname of the database server,
- `port`: port on the database server. It must not contain a password component. Components that are not applicable should be set to NA.

For objects of class [DBI::DBIResult](#), `dbGetInfo()` returns a named list that contains at least the following components:

- `statatment`: the statement used with [DBI::dbSendQuery\(\)](#) or [DBI::dbExecute\(\)](#), as returned by [DBI::dbGetStatement\(\)](#),
- `row.count`: the number of rows fetched so far (for queries), as returned by [DBI::dbGetRowCount\(\)](#),
- `rows.affected`: the number of rows affected (for statements), as returned by [DBI::dbGetRowsAffected\(\)](#)
- `has.completed`: a logical that indicates if the query or statement has completed, as returned by [DBI::dbHasCompleted\(\)](#).

See Also

Other DBIDriver generics: [DBIDriver-class](#), [dbCanConnect\(\)](#), [dbConnect\(\)](#), [dbDataType\(\)](#), [dbDriver\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListConnections\(\)](#)

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

dbGetInfo, DatabaseConnectorDriver-method

Get DBMS metadata

Description

Retrieves information on objects of class [DBIDriver](#), [DBIConnection](#) or [DBIResult](#).

Usage

```
## S4 method for signature 'DatabaseConnectorDriver'
dbGetInfo(dbObj, ...)
```

Arguments

dbObj	An object inheriting from DBIObject , i.e. DBIDriver , DBIConnection , or a DBIResult
...	Other arguments to methods.

Value

For objects of class [DBI::DBIDriver](#), [dbGetInfo\(\)](#) returns a named list that contains at least the following components:

- `driver.version`: the package version of the DBI backend,
- `client.version`: the version of the DBMS client library.

For objects of class [DBI::DBIConnection](#), [dbGetInfo\(\)](#) returns a named list that contains at least the following components:

- `db.version`: version of the database server,
- `dbname`: database name,
- `username`: username to connect to the database,
- `host`: hostname of the database server,
- `port`: port on the database server. It must not contain a password component. Components that are not applicable should be set to NA.

For objects of class [DBI::DBIResult](#), `dbGetInfo()` returns a named list that contains at least the following components:

- `statement`: the statement used with [DBI::dbSendQuery\(\)](#) or [DBI::dbExecute\(\)](#), as returned by [DBI::dbGetStatement\(\)](#),
- `row.count`: the number of rows fetched so far (for queries), as returned by [DBI::dbGetRowCount\(\)](#),
- `rows.affected`: the number of rows affected (for statements), as returned by [DBI::dbGetRowsAffected\(\)](#)
- `has.completed`: a logical that indicates if the query or statement has completed, as returned by [DBI::dbHasCompleted\(\)](#).

See Also

Other DBIDriver generics: [DBIDriver-class](#), [dbCanConnect\(\)](#), [dbConnect\(\)](#), [dbDataType\(\)](#), [dbDriver\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListConnections\(\)](#)

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

dbGetQuery, DatabaseConnectorDbiConnection, character-method

Retrieve results from a query

Description

Returns the result of a query as a data frame. `dbGetQuery()` comes with a default implementation (which should work with most backends) that calls [dbSendQuery\(\)](#), then [dbFetch\(\)](#), ensuring that the result is always freed by [dbClearResult\(\)](#). For retrieving chunked/paged results or for passing query parameters, see [dbSendQuery\(\)](#), in particular the "The data retrieval flow" section. For retrieving results as an Arrow object, see [dbGetQueryArrow\(\)](#).

Usage

```
## S4 method for signature 'DatabaseConnectorDbiConnection,character'
dbGetQuery(conn, statement, ...)
```

Arguments

<code>conn</code>	A DBI::DBIConnection object, as returned by dbConnect() .
<code>statement</code>	a character string containing SQL.
<code>...</code>	Other parameters passed on to methods.

Details

This method is for SELECT queries only (incl. other SQL statements that return a SELECT-alike result, e.g., execution of a stored procedure or data manipulation queries like INSERT INTO ... RETURNING ...). To execute a stored procedure that does not return a result set, use [dbExecute\(\)](#).

Some backends may support data manipulation statements through this method for compatibility reasons. However, callers are strongly advised to use [dbExecute\(\)](#) for data manipulation statements.

Value

dbGetQuery() always returns a [data.frame](#), with as many rows as records were fetched and as many columns as fields in the result set, even if the result is a single value or has one or zero rows.

See Also

For updates: [dbSendStatement\(\)](#) and [dbExecute\(\)](#).

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetch\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQueryArrow\(\)](#), [dbHasCompleted\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#)

dbGetQuery, DatabaseConnectorJdbcConnection, character-method

Retrieve results from a query

Description

Returns the result of a query as a data frame. dbGetQuery() comes with a default implementation (which should work with most backends) that calls [dbSendQuery\(\)](#), then [dbFetch\(\)](#), ensuring that the result is always freed by [dbClearResult\(\)](#). For retrieving chunked/paged results or for passing query parameters, see [dbSendQuery\(\)](#), in particular the "The data retrieval flow" section. For retrieving results as an Arrow object, see [dbGetQueryArrow\(\)](#).

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcConnection,character'
dbGetQuery(conn, statement, ...)
```

Arguments

conn	A DBI::DBIConnection object, as returned by dbConnect() .
statement	a character string containing SQL.
...	Other parameters passed on to methods.

Details

This method is for SELECT queries only (incl. other SQL statements that return a SELECT-alike result, e.g., execution of a stored procedure or data manipulation queries like INSERT INTO ... RETURNING ...). To execute a stored procedure that does not return a result set, use [dbExecute\(\)](#).

Some backends may support data manipulation statements through this method for compatibility reasons. However, callers are strongly advised to use [dbExecute\(\)](#) for data manipulation statements.

Value

dbGetQuery() always returns a [data.frame](#), with as many rows as records were fetched and as many columns as fields in the result set, even if the result is a single value or has one or zero rows.

See Also

For updates: [dbSendStatement\(\)](#) and [dbExecute\(\)](#).

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetch\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQueryArrow\(\)](#), [dbHasCompleted\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#)

dbGetRowCount, DatabaseConnectorDbiResult-method

The number of rows fetched so far

Description

Returns the total number of rows actually fetched with calls to [dbFetch\(\)](#) for this result set.

Usage

```
## S4 method for signature 'DatabaseConnectorDbiResult'
dbGetRowCount(res, ...)
```

Arguments

res	An object inheriting from DBI::DBIResult .
...	Other arguments passed on to methods.

Value

dbGetRowCount() returns a scalar number (integer or numeric), the number of rows fetched so far. After calling [DBI::dbSendQuery\(\)](#), the row count is initially zero. After a call to [DBI::dbFetch\(\)](#) without limit, the row count matches the total number of rows returned. Fetching a limited number of rows increases the number of rows by the number of rows returned, even if fetching past the end of the result set. For queries with an empty result set, zero is returned even after fetching. For data manipulation statements issued with [DBI::dbSendStatement\(\)](#), zero is returned before and after calling [dbFetch\(\)](#).

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

dbGetRowCount, DatabaseConnectorJdbcResult-method

The number of rows fetched so far

Description

Returns the total number of rows actually fetched with calls to [dbFetch\(\)](#) for this result set.

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcResult'
dbGetRowCount(res, ...)
```

Arguments

res	An object inheriting from DBI::DBIResult .
...	Other arguments passed on to methods.

Value

`dbGetRowCount()` returns a scalar number (integer or numeric), the number of rows fetched so far. After calling [DBI::dbSendQuery\(\)](#), the row count is initially zero. After a call to [DBI::dbFetch\(\)](#) without limit, the row count matches the total number of rows returned. Fetching a limited number of rows increases the number of rows by the number of rows returned, even if fetching past the end of the result set. For queries with an empty result set, zero is returned even after fetching. For data manipulation statements issued with [DBI::dbSendStatement\(\)](#), zero is returned before and after calling `dbFetch()`.

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

dbGetRowsAffected, DatabaseConnectorDbiResult-method

The number of rows affected

Description

This method returns the number of rows that were added, deleted, or updated by a data manipulation statement.

Usage

```
## S4 method for signature 'DatabaseConnectorDbiResult'
dbGetRowsAffected(res, ...)
```

Arguments

res	An object inheriting from DBI::DBIResult .
...	Other arguments passed on to methods.

Value

dbGetRowsAffected() returns a scalar number (integer or numeric), the number of rows affected by a data manipulation statement issued with [DBI::dbSendStatement\(\)](#). The value is available directly after the call and does not change after calling [DBI::dbFetch\(\)](#). NA_integer_ or NA_numeric_ are allowed if the number of rows affected is not known.

For queries issued with [DBI::dbSendQuery\(\)](#), zero is returned before and after the call to dbFetch(). NA values are not allowed.

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other command execution generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbExecute\(\)](#), [dbSendStatement\(\)](#)

dbGetRowsAffected, DatabaseConnectorJdbcResult-method

The number of rows affected

Description

This method returns the number of rows that were added, deleted, or updated by a data manipulation statement.

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcResult'
dbGetRowsAffected(res, ...)
```

Arguments

res An object inheriting from [DBI::DBIResult](#).
 ... Other arguments passed on to methods.

Value

dbGetRowsAffected() returns a scalar number (integer or numeric), the number of rows affected by a data manipulation statement issued with [DBI::dbSendStatement\(\)](#). The value is available directly after the call and does not change after calling [DBI::dbFetch\(\)](#). NA_integer_ or NA_numeric_ are allowed if the number of rows affected is not known.

For queries issued with [DBI::dbSendQuery\(\)](#), zero is returned before and after the call to [dbFetch\(\)](#). NA values are not allowed.

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other command execution generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbExecute\(\)](#), [dbSendStatement\(\)](#)

dbGetStatement, DatabaseConnectorDbiResult-method

Get the statement associated with a result set

Description

Returns the statement that was passed to [dbSendQuery\(\)](#) or [dbSendStatement\(\)](#).

Usage

```
## S4 method for signature 'DatabaseConnectorDbiResult'
dbGetStatement(res, ...)
```

Arguments

res An object inheriting from [DBI::DBIResult](#).
 ... Other arguments passed on to methods.

Value

dbGetStatement() returns a string, the query used in either [DBI::dbSendQuery\(\)](#) or [DBI::dbSendStatement\(\)](#).

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

dbGetStatement, DatabaseConnectorJdbcResult-method
Get the statement associated with a result set

Description

Returns the statement that was passed to [dbSendQuery\(\)](#) or [dbSendStatement\(\)](#).

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcResult'
dbGetStatement(res, ...)
```

Arguments

res	An object inheriting from DBI::DBIResult .
...	Other arguments passed on to methods.

Value

[dbGetStatement\(\)](#) returns a string, the query used in either [DBI::dbSendQuery\(\)](#) or [DBI::dbSendStatement\(\)](#).

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

dbHasCompleted, DatabaseConnectorDbiResult-method
Completion status

Description

This method returns if the operation has completed. A SELECT query is completed if all rows have been fetched. A data manipulation statement is always completed.

Usage

```
## S4 method for signature 'DatabaseConnectorDbiResult'
dbHasCompleted(res, ...)
```

Arguments

res	An object inheriting from DBI::DBIResult .
...	Other arguments passed on to methods.

Value

dbHasCompleted() returns a logical scalar. For a query initiated by [DBI::dbSendQuery\(\)](#) with non-empty result set, dbHasCompleted() returns FALSE initially and TRUE after calling [DBI::dbFetch\(\)](#) without limit. For a query initiated by [DBI::dbSendStatement\(\)](#), dbHasCompleted() always returns TRUE.

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other DBIResultArrow generics: [DBIResultArrow-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbIsValid\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetch\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#)

dbHasCompleted, DatabaseConnectorJdbcResult-method

Completion status

Description

This method returns if the operation has completed. A SELECT query is completed if all rows have been fetched. A data manipulation statement is always completed.

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcResult'
dbHasCompleted(res, ...)
```

Arguments

res	An object inheriting from DBI::DBIResult .
...	Other arguments passed on to methods.

Value

dbHasCompleted() returns a logical scalar. For a query initiated by [DBI::dbSendQuery\(\)](#) with non-empty result set, dbHasCompleted() returns FALSE initially and TRUE after calling [DBI::dbFetch\(\)](#) without limit. For a query initiated by [DBI::dbSendStatement\(\)](#), dbHasCompleted() always returns TRUE.

See Also

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other DBIResultArrow generics: [DBIResultArrow-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbIsValid\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetch\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#)

dbIsValid, DatabaseConnectorDbiConnection-method
Is this DBMS object still valid?

Description

This generic tests whether a database object is still valid (i.e. it hasn't been disconnected or cleared).

Usage

```
## S4 method for signature 'DatabaseConnectorDbiConnection'
dbIsValid(dbObj, ...)
```

Arguments

dbObj	An object inheriting from DBIObject , i.e. DBIDriver , DBIConnection , or a DBIResult
...	Other arguments to methods.

Value

dbIsValid() returns a logical scalar, TRUE if the object specified by dbObj is valid, FALSE otherwise. A [DBI::DBIConnection](#) object is initially valid, and becomes invalid after disconnecting with [DBI::dbDisconnect\(\)](#). For an invalid connection object (e.g., for some drivers if the object is saved to a file and then restored), the method also returns FALSE. A [DBI::DBIResult](#) object is valid after a call to [DBI::dbSendQuery\(\)](#), and stays valid even after all rows have been fetched; only clearing it with [DBI::dbClearResult\(\)](#) invalidates it. A [DBI::DBIResult](#) object is also valid after a call to [DBI::dbSendStatement\(\)](#), and stays valid after querying the number of rows affected; only clearing it with [DBI::dbClearResult\(\)](#) invalidates it. If the connection to the database system is dropped (e.g., due to connectivity problems, server failure, etc.), dbIsValid() should return FALSE. This is not tested automatically.

See Also

Other DBIDriver generics: [DBIDriver-class](#), [dbCanConnect\(\)](#), [dbConnect\(\)](#), [dbDataType\(\)](#), [dbDriver\(\)](#), [dbGetInfo\(\)](#), [dbIsReadOnly\(\)](#), [dbListConnections\(\)](#)

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other DBIResultArrow generics: [DBIResultArrow-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbHasCompleted\(\)](#)

dbIsValid, DatabaseConnectorJdbcConnection-method

Is this DBMS object still valid?

Description

This generic tests whether a database object is still valid (i.e. it hasn't been disconnected or cleared).

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcConnection'
dbIsValid(dbObj, ...)
```

Arguments

dbObj	An object inheriting from DBIObject , i.e. DBIDriver , DBIConnection , or a DBIResult
...	Other arguments to methods.

Value

dbIsValid() returns a logical scalar, TRUE if the object specified by dbObj is valid, FALSE otherwise. A [DBI::DBIConnection](#) object is initially valid, and becomes invalid after disconnecting with [DBI::dbDisconnect\(\)](#). For an invalid connection object (e.g., for some drivers if the object is saved to a file and then restored), the method also returns FALSE. A [DBI::DBIResult](#) object is valid after a call to [DBI::dbSendQuery\(\)](#), and stays valid even after all rows have been fetched; only clearing it with [DBI::dbClearResult\(\)](#) invalidates it. A [DBI::DBIResult](#) object is also valid after a call to [DBI::dbSendStatement\(\)](#), and stays valid after querying the number of rows affected; only clearing it with [DBI::dbClearResult\(\)](#) invalidates it. If the connection to the database system is dropped (e.g., due to connectivity problems, server failure, etc.), dbIsValid() should return FALSE. This is not tested automatically.

See Also

Other DBIDriver generics: [DBIDriver-class](#), [dbCanConnect\(\)](#), [dbConnect\(\)](#), [dbDataType\(\)](#), [dbDriver\(\)](#), [dbGetInfo\(\)](#), [dbIsReadOnly\(\)](#), [dbListConnections\(\)](#)

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other DBIResult generics: [DBIResult-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbColumnInfo\(\)](#), [dbFetch\(\)](#), [dbGetInfo\(\)](#), [dbGetRowCount\(\)](#), [dbGetRowsAffected\(\)](#), [dbGetStatement\(\)](#), [dbHasCompleted\(\)](#), [dbIsReadOnly\(\)](#), [dbQuoteLiteral\(\)](#), [dbQuoteString\(\)](#)

Other DBIResultArrow generics: [DBIResultArrow-class](#), [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbHasCompleted\(\)](#)

dbListFields, DatabaseConnectorConnection, character-method
List field names of a remote table

Description

Returns the field names of a remote table as a character vector.

Usage

```
## S4 method for signature 'DatabaseConnectorConnection,character'
dbListFields(conn, name, databaseSchema = NULL, ...)
```

Arguments

conn	A DBI::DBIConnection object, as returned by dbConnect() .
name	The table name, passed on to dbQuoteIdentifier() . Options are: • a character string with the unquoted DBMS table name, e.g. "table_name", • a call to Id() with components to the fully qualified table name, e.g. Id(schema = "my_schema", table = "table_name") • a call to SQL() with the quoted and fully qualified table name given verbatim, e.g. SQL('"my_schema"."table_name"')
databaseSchema	The name of the database schema. See details for platform-specific details.
...	Other parameters passed on to methods.

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

dbListFields() returns a character vector that enumerates all fields in the table in the correct order. This also works for temporary tables if supported by the database. The returned names are suitable for quoting with dbQuoteIdentifier().

See Also

[dbColumnInfo\(\)](#) to get the type of the fields.

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

dbListTables, DatabaseConnectorConnection-method

List remote tables

Description

Returns the unquoted names of remote tables accessible through this connection. This should include views and temporary objects, but not all database backends (in particular **RMariaDB** and **RMySQL**) support this.

Usage

```
## S4 method for signature 'DatabaseConnectorConnection'
dbListTables(conn, databaseSchema = NULL, ...)
```

Arguments

conn	A DBI::DBIConnection object, as returned by dbConnect() .
databaseSchema	The name of the database schema. See details for platform-specific details.
...	Not used

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

dbListTables() returns a character vector that enumerates all tables and views in the database. Tables added with [DBI::dbWriteTable\(\)](#) are part of the list. As soon a table is removed from the database, it is also removed from the list of database tables.

The same applies to temporary tables if supported by the database.

The returned names are suitable for quoting with [dbQuoteIdentifier\(\)](#).

See Also

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

dbms	<i>Get the database platform from a connection</i>
------	--

Description

The SqlRender package provides functions that translate SQL from OHDSI-SQL to a target SQL dialect. These function need the name of the database platform to translate to. The dbms function returns the dbms for any DBI connection that can be passed along to SqlRender translation functions (see example).

Usage

```
dbms(connection)
```

Arguments

connection The connection to the database server created using either `connect()` or `DBI::dbConnect()`.

Value

The name of the database (dbms) used by SqlRender

Examples

```
library(DatabaseConnector)
con <- connect(dbms = "sqlite", server = ":memory:")
dbms(con)
#> [1] "sqlite"
SqlRender::translate("DATEADD(d, 365, dateColumn)", targetDialect = dbms(con))
#> "CAST(STRFTIME('%s', DATETIME(dateColumn, 'unixepoch', (365)||' days')) AS REAL)"
disconnect(con)
```

dbplyr_edition.DatabaseConnectorConnection	<i>Declare dbplyr backend edition</i>
--	---------------------------------------

Description

Declare dbplyr backend edition

Usage

```
## S3 method for class 'DatabaseConnectorConnection'
dbplyr_edition(con)

## S3 method for class 'DatabaseConnectorJdbcConnection'
dbplyr_edition(con)

## S3 method for class 'DatabaseConnectorDbiConnection'
dbplyr_edition(con)
```

Arguments

con Database connection

dbReadTable, DatabaseConnectorConnection, character-method

Read database tables as data frames

Description

Reads a database table to a data frame, optionally converting a column to row names and converting the column names to valid R identifiers. Use `dbReadTableArrow()` instead to obtain an Arrow object.

Usage

```
## S4 method for signature 'DatabaseConnectorConnection,character'
dbReadTable(conn, name, databaseSchema = NULL, ...)
```

Arguments

conn A [DBI::DBIConnection](#) object, as returned by `dbConnect()`.

name The table name, passed on to `dbQuoteIdentifier()`. Options are:

- a character string with the unquoted DBMS table name, e.g. "table_name",
- a call to `Id()` with components to the fully qualified table name, e.g. `Id(schema = "my_schema", table = "table_name")`
- a call to `SQL()` with the quoted and fully qualified table name given verbatim, e.g. `SQL('"my_schema"."table_name"')`

databaseSchema The name of the database schema. See details for platform-specific details.

... Other parameters passed on to methods.

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

`dbReadTable()` returns a data frame that contains the complete data from the remote table, effectively the result of calling `DBI::dbGetQuery()` with `SELECT * FROM <name>`.

An empty table is returned as a data frame with zero rows.

The presence of `rownames` depends on the `row.names` argument, see `DBI::sqlColumnToRownames()` for details:

- If FALSE or NULL, the returned data frame doesn't have row names.
- If TRUE, a column named "row_names" is converted to row names.

- If NA, a column named "row_names" is converted to row names if it exists, otherwise no translation occurs.
- If a string, this specifies the name of the column in the remote table that contains the row names.

The default is `row.names = FALSE`.

If the database supports identifiers with special characters, the columns in the returned data frame are converted to valid R identifiers if the `check.names` argument is `TRUE`. If `check.names = FALSE`, the returned table has non-syntactic column names without quotes.

See Also

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

dbRemoveTable, DatabaseConnectorConnection, ANY-method

Remove a table from the database

Description

Remove a remote table (e.g., created by [dbWriteTable\(\)](#)) from the database.

Usage

```
## S4 method for signature 'DatabaseConnectorConnection,ANY'
dbRemoveTable(conn, name, databaseSchema = NULL, ...)
```

Arguments

<code>conn</code>	A DBI::DBIConnection object, as returned by dbConnect() .
<code>name</code>	The table name, passed on to dbQuoteIdentifier() . Options are: • a character string with the unquoted DBMS table name, e.g. <code>"table_name"</code>, • a call to Id() with components to the fully qualified table name, e.g. <code>Id(schema = "my_schema", table = "table_name")</code> • a call to SQL() with the quoted and fully qualified table name given verbatim, e.g. <code>SQL('"my_schema"."table_name"')</code>
<code>databaseSchema</code>	The name of the database schema. See details for platform-specific details.
<code>...</code>	Other parameters passed on to methods.

Details

The `databaseSchema` argument is interpreted differently according to the different platforms: SQL Server and PDW: The `databaseSchema` schema should specify both the database and the schema, e.g. `'my_database.dbo'`. Impala: the `databaseSchema` should specify the database. Oracle: The `databaseSchema` should specify the Oracle 'user'. All other : The `databaseSchema` should specify the schema.

Value

dbRemoveTable() returns TRUE, invisibly.

See Also

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

dbSendQuery, DatabaseConnectorDbiConnection, character-method

Execute a query on a given database connection

Description

The dbSendQuery() method only submits and synchronously executes the SQL query to the database engine. It does *not* extract any records — for that you need to use the [dbFetch\(\)](#) method, and then you must call [dbClearResult\(\)](#) when you finish fetching the records you need. For interactive use, you should almost always prefer [dbGetQuery\(\)](#). Use [dbSendQueryArrow\(\)](#) or [dbGetQueryArrow\(\)](#) instead to retrieve the results as an Arrow object.

Usage

```
## S4 method for signature 'DatabaseConnectorDbiConnection,character'
dbSendQuery(conn, statement, ...)
```

Arguments

conn	A DBI::DBIConnection object, as returned by dbConnect() .
statement	a character string containing SQL.
...	Other parameters passed on to methods.

Details

This method is for SELECT queries only. Some backends may support data manipulation queries through this method for compatibility reasons. However, callers are strongly encouraged to use [dbSendStatement\(\)](#) for data manipulation statements.

The query is submitted to the database server and the DBMS executes it, possibly generating vast amounts of data. Where these data live is driver-specific: some drivers may choose to leave the output on the server and transfer them piecemeal to R, others may transfer all the data to the client – but not necessarily to the memory that R manages. See individual drivers' dbSendQuery() documentation for details.

Value

dbSendQuery() returns an S4 object that inherits from [DBI::DBIResult](#). The result set can be used with [DBI::dbFetch\(\)](#) to extract records. Once you have finished using a result, make sure to clear it with [DBI::dbClearResult\(\)](#).

See Also

For updates: [dbSendStatement\(\)](#) and [dbExecute\(\)](#).

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetch\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbHasCompleted\(\)](#), [dbSendQueryArrow\(\)](#)

dbSendQuery, DatabaseConnectorJdbcConnection, character-method

Execute a query on a given database connection

Description

The `dbSendQuery()` method only submits and synchronously executes the SQL query to the database engine. It does *not* extract any records — for that you need to use the [dbFetch\(\)](#) method, and then you must call [dbClearResult\(\)](#) when you finish fetching the records you need. For interactive use, you should almost always prefer [dbGetQuery\(\)](#). Use [dbSendQueryArrow\(\)](#) or [dbGetQueryArrow\(\)](#) instead to retrieve the results as an Arrow object.

Usage

```
## S4 method for signature 'DatabaseConnectorJdbcConnection,character'
dbSendQuery(conn, statement, ...)
```

Arguments

<code>conn</code>	A DBI::DBIConnection object, as returned by dbConnect() .
<code>statement</code>	a character string containing SQL.
<code>...</code>	Other parameters passed on to methods.

Details

This method is for SELECT queries only. Some backends may support data manipulation queries through this method for compatibility reasons. However, callers are strongly encouraged to use [dbSendStatement\(\)](#) for data manipulation statements.

The query is submitted to the database server and the DBMS executes it, possibly generating vast amounts of data. Where these data live is driver-specific: some drivers may choose to leave the output on the server and transfer them piecemeal to R, others may transfer all the data to the client – but not necessarily to the memory that R manages. See individual drivers' `dbSendQuery()` documentation for details.

Value

`dbSendQuery()` returns an S4 object that inherits from [DBI::DBIResult](#). The result set can be used with [DBI::dbFetch\(\)](#) to extract records. Once you have finished using a result, make sure to clear it with [DBI::dbClearResult\(\)](#).

See Also

For updates: [dbSendStatement\(\)](#) and [dbExecute\(\)](#).

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other data retrieval generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbFetch\(\)](#), [dbFetchArrow\(\)](#), [dbFetchArrowChunk\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbHasCompleted\(\)](#), [dbSendQueryArrow\(\)](#)

dbSendStatement, DatabaseConnectorConnection, character-method

Execute a data manipulation statement on a given database connection

Description

The `dbSendStatement()` method only submits and synchronously executes the SQL data manipulation statement (e.g., UPDATE, DELETE, INSERT INTO, DROP TABLE, ...) to the database engine. To query the number of affected rows, call [dbGetRowsAffected\(\)](#) on the returned result object. You must also call [dbClearResult\(\)](#) after that. For interactive use, you should almost always prefer [dbExecute\(\)](#).

Usage

```
## S4 method for signature 'DatabaseConnectorConnection,character'
dbSendStatement(conn, statement, ...)
```

Arguments

<code>conn</code>	A DBI::DBIConnection object, as returned by dbConnect() .
<code>statement</code>	a character string containing SQL.
<code>...</code>	Other parameters passed on to methods.

Details

[dbSendStatement\(\)](#) comes with a default implementation that simply forwards to [dbSendQuery\(\)](#), to support backends that only implement the latter.

Value

`dbSendStatement()` returns an S4 object that inherits from [DBI::DBIResult](#). The result set can be used with [DBI::dbGetRowsAffected\(\)](#) to determine the number of rows affected by the query. Once you have finished using a result, make sure to clear it with [DBI::dbClearResult\(\)](#).

See Also

For queries: [dbSendQuery\(\)](#) and [dbGetQuery\(\)](#).

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTable\(\)](#), [dbWriteTableArrow\(\)](#)

Other command execution generics: [dbBind\(\)](#), [dbClearResult\(\)](#), [dbExecute\(\)](#), [dbGetRowsAffected\(\)](#)

dbUnloadDriver, DatabaseConnectorDriver-method

Load and unload database drivers

Description

These methods are deprecated, please consult the documentation of the individual backends for the construction of driver instances.

`dbDriver()` is a helper method used to create a new driver object given the name of a database or the corresponding R package. It works through convention: all DBI-extending packages should provide an exported object with the same name as the package. `dbDriver()` just looks for this object in the right places: if you know what database you are connecting to, you should call the function directly.

`dbUnloadDriver()` is not implemented for modern backends.

Usage

```
## S4 method for signature 'DatabaseConnectorDriver'
dbUnloadDriver(drv, ...)
```

Arguments

<code>drv</code>	an object that inherits from <code>DBIDriver</code> as created by <code>dbDriver</code> .
<code>...</code>	any other arguments are passed to the driver <code>drvName</code> .

Details

The client part of the database communication is initialized (typically dynamically loading C code, etc.) but note that connecting to the database engine itself needs to be done through calls to `dbConnect`.

Value

In the case of `dbDriver`, an driver object whose class extends `DBIDriver`. This object may be used to create connections to the actual DBMS engine.

In the case of `dbUnloadDriver`, a logical indicating whether the operation succeeded or not.

See Also

Other DBIDriver generics: [DBIDriver-class](#), [dbCanConnect\(\)](#), [dbConnect\(\)](#), [dbDataType\(\)](#), [dbGetInfo\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListConnections\(\)](#)

Other DBIDriver generics: [DBIDriver-class](#), [dbCanConnect\(\)](#), [dbConnect\(\)](#), [dbDataType\(\)](#), [dbGetInfo\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListConnections\(\)](#)

dbWriteTable, DatabaseConnectorConnection, ANY-method
Copy data frames to database tables

Description

Writes, overwrites or appends a data frame to a database table, optionally converting row names to a column and specifying SQL data types for fields.

Usage

```
## S4 method for signature 'DatabaseConnectorConnection,ANY'
dbWriteTable(
  conn,
  name,
  value,
  databaseSchema = NULL,
  overwrite = FALSE,
  append = FALSE,
  temporary = FALSE,
  ...
)
```

Arguments

conn	A DBI::DBIConnection object, as returned by dbConnect() .
name	The table name, passed on to dbQuoteIdentifier() . Options are: • a character string with the unquoted DBMS table name, e.g. "table_name", • a call to Id() with components to the fully qualified table name, e.g. Id(schema = "my_schema", table = "table_name") • a call to SQL() with the quoted and fully qualified table name given verbatim, e.g. SQL('"my_schema"."table_name"')
value	A data.frame (or coercible to data.frame).
databaseSchema	The name of the database schema. See details for platform-specific details.
overwrite	Overwrite an existing table (if exists)?
append	Append to existing table?
temporary	Should the table created as a temp table?
...	Other parameters passed on to methods.

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

dbWriteTable() returns TRUE, invisibly.

See Also

Other DBIConnection generics: [DBIConnection-class](#), [dbAppendTable\(\)](#), [dbAppendTableArrow\(\)](#), [dbCreateTable\(\)](#), [dbCreateTableArrow\(\)](#), [dbDataType\(\)](#), [dbDisconnect\(\)](#), [dbExecute\(\)](#), [dbExistsTable\(\)](#), [dbGetException\(\)](#), [dbGetInfo\(\)](#), [dbGetQuery\(\)](#), [dbGetQueryArrow\(\)](#), [dbIsReadOnly\(\)](#), [dbIsValid\(\)](#), [dbListFields\(\)](#), [dbListObjects\(\)](#), [dbListResults\(\)](#), [dbListTables\(\)](#), [dbQuoteIdentifier\(\)](#), [dbReadTable\(\)](#), [dbReadTableArrow\(\)](#), [dbRemoveTable\(\)](#), [dbSendQuery\(\)](#), [dbSendQueryArrow\(\)](#), [dbSendStatement\(\)](#), [dbUnquoteIdentifier\(\)](#), [dbWriteTableArrow\(\)](#)

disconnect

Disconnect from the server

Description

Close the connection to the server.

Usage

```
disconnect(connection)
```

Arguments

connection The connection to the database server created using either [connect\(\)](#) or [DBI::dbConnect\(\)](#).

Examples

```
## Not run:
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = "localhost",
  user = "root",
  password = "blah"
)
conn <- connect(connectionDetails)
count <- querySql(conn, "SELECT COUNT(*) FROM person")
disconnect(conn)

## End(Not run)
```

downloadJdbcDrivers *Download DatabaseConnector JDBC Jar files*

Description

Download the DatabaseConnector JDBC drivers from <https://ohdsi.github.io/DatabaseConnectorJars/>

Usage

```
downloadJdbcDrivers(
  dbms,
  pathToDriver = Sys.getenv("DATABASECONNECTOR_JAR_FOLDER"),
  method = "auto",
  ...
)
```

Arguments

dbms	The type of DBMS to download Jar files for. • "postgresql" for PostgreSQL • "redshift" for Amazon Redshift • "sql server", "pdw" or "synapse" for Microsoft SQL Server • "oracle" for Oracle • "spark" for Spark (Databricks) • "snowflake" for Snowflake • "bigquery" for Google BigQuery • "iris" for InterSystems IRIS • "all" for all aforementioned platforms
pathToDriver	The full path to the folder where the JDBC driver .jar files should be downloaded to. By default the value of the environment variable "DATABASECONNECTOR_JAR_FOLDER" is used.
method	The method used for downloading files. See ?download.file for details and options.
...	Further arguments passed on to download.file.

Details

The following versions of the JDBC drivers are currently used:

- PostgreSQL: V42.7.3
- RedShift: V2.1.0.20
- SQL Server: V9.2.0
- Oracle: V19.8
- Spark (Databricks): V2.6.36
- Snowflake: V4.3.4
- BigQuery: v0.10.0
- InterSystems IRIS: v3.11.0

Value

Invisibly returns the destination if the download was successful.

Examples

```
## Not run:
downloadJdbcDrivers("redshift")

## End(Not run)
```

dropEmulatedTempTables

Drop all emulated temp tables.

Description

On some DBMSs, like Oracle and BigQuery, DatabaseConnector through SqlRender emulates temp tables in a schema provided by the user. Ideally, these tables are deleted by the application / R script creating them, but for various reasons orphan temp tables may remain. This function drops all emulated temp tables created in this session only.

Usage

```
dropEmulatedTempTables(
  connection,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema")
)
```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

Value

Invisibly returns the list of deleted emulated temp tables.

executeSql

*Execute SQL code***Description**

This function executes SQL consisting of one or more statements.

Usage

```
executeSql(
  connection,
  sql,
  profile = FALSE,
  progressBar = !as.logical(Sys.getenv("TESTTHAT", unset = FALSE)),
  reportOverallTime = TRUE,
  errorReportFile = file.path(getOption("errorReportFileLocation", getwd()),
    "errorReportSql.txt"),
  runAsBatch = FALSE
)
```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
sql	The SQL to be executed.
profile	When true, each separate statement is written to file prior to sending to the server, and the time taken to execute a statement is displayed.
progressBar	When true, a progress bar is shown based on the statements in the SQL code.
reportOverallTime	When true, the function will display the overall time taken to execute all statements.
errorReportFile	The file where an error report will be written if an error occurs. Defaults to 'errorReportSql.txt' in the directory given by <code>getOption("errorReportFileLocation")</code> , or the current working directory if that option is unset.
runAsBatch	When true the SQL statements are sent to the server as a single batch, and executed there. This will be faster if you have many small SQL statements, but there will be no progress bar, and no per-statement error messages. If the database platform does not support batched updates the query is executed without batching.

Details

This function splits the SQL in separate statements and sends it to the server for execution. If an error occurs during SQL execution, this error is written to a file to facilitate debugging. Optionally, a progress bar is shown and the total time taken to execute the SQL is displayed. Optionally, each separate SQL statement is written to file, and the execution time per statement is shown to aid in detecting performance issues.

Examples

```
## Not run:
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = "localhost",
  user = "root",
  password = "blah",
  schema = "cdm_v4"
)
conn <- connect(connectionDetails)
executeSql(conn, "CREATE TABLE x (k INT); CREATE TABLE y (k INT);")
disconnect(conn)

## End(Not run)
```

existsTable	<i>Does the table exist?</i>
-------------	------------------------------

Description

Checks whether a table exists. Accounts for surrounding escape characters. Case insensitive.

Usage

```
existsTable(connection, databaseSchema, tableName)
```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
databaseSchema	The name of the database schema. See details for platform-specific details.
tableName	The name of the table to check.

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

A logical value indicating whether the table exists.

extractQueryTimes	<i>Extract query times from a ParallelLogger log file</i>
-------------------	---

Description

When using the ParallelLogger default file logger, and using options(LOG_DATABASECONNECTOR_SQL = TRUE), DatabaseConnector will log all SQL sent to the server, and the time to get a response.

This function parses the log file, producing a data frame with time per query.

Usage

```
extractQueryTimes(logFileName)
```

Arguments

logFileName	Name of the ParallelLogger log file. Assumes the file was created using the default file logger.
-------------	--

Value

A data frame with queries and their run times in milliseconds.

Examples

```
connection <- connect(dbms = "sqlite", server = ":memory:")
logFile <- tempfile(fileext = ".log")
ParallelLogger::addDefaultFileLogger(fileName = logFile, name = "MY_LOGGER")
options(LOG_DATABASECONNECTOR_SQL = TRUE)

executeSql(connection, "CREATE TABLE test (x INT);")
querySql(connection, "SELECT * FROM test;")

extractQueryTimes(logFile)

ParallelLogger::unregisterLogger("MY_LOGGER")
unlink(logFile)
disconnect(connection)
```

getAvailableJavaHeapSpace	<i>Get available Java heap space</i>
---------------------------	--------------------------------------

Description

For debugging purposes: get the available Java heap space.

Usage

```
getAvailableJavaHeapSpace()
```

Value

The Java heap space (in bytes).

getTableNames	<i>List all tables in a database schema.</i>
---------------	--

Description

This function returns a list of all tables in a database schema.

Usage

```
getTableNames(connection, databaseSchema = NULL, cast = "lower")
```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
databaseSchema	The name of the database schema. See details for platform-specific details.
cast	Should the table names be cast to uppercase or lowercase before being returned? Valid options are "upper" , "lower" (default), "none" (no casting is done)

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

A character vector of table names.

inDatabaseSchema	<i>Refer to a table in a database schema</i>
------------------	--

Description

Can be used with `dplyr::tbl()` to indicate a table in a specific database schema.

Usage

```
inDatabaseSchema(databaseSchema, table)
```

Arguments

databaseSchema	The name of the database schema. See details for platform-specific details.
table	The name of the table in the database schema.

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

Value

An object representing the table and database schema.

insertTable	<i>Insert a table on the server</i>
-------------	-------------------------------------

Description

This function sends the data in a data frame to a table on the server. Either a new table is created, or the data is appended to an existing table.

Usage

```
insertTable(
  connection,
  databaseSchema = NULL,
  tableName,
  data,
  dropTableIfExists = TRUE,
  createTable = TRUE,
  tempTable = FALSE,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  bulkLoad = Sys.getenv("DATABASE_CONNECTOR_BULK_UPLOAD"),
  useMppBulkLoad = Sys.getenv("USE_MPP_BULK_LOAD"),
  progressBar = FALSE,
  camelCaseToSnakeCase = FALSE
)
```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
databaseSchema	The name of the database schema. See details for platform-specific details.
tableName	The name of the table where the data should be inserted.
data	The data frame containing the data to be inserted.
dropTableIfExists	Drop the table if the table already exists before writing?
createTable	Create a new table? If false, will append to existing table.
tempTable	Should the table created as a temp table?
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

bulkLoad	If using Redshift, PDW, Hive or Postgres, use more performant bulk loading techniques. Does not work for temp tables (except for HIVE). See Details for requirements for the various platforms.
useMppBulkLoad	DEPRECATED. Use bulkLoad instead.
progressBar	Show a progress bar when uploading?
camelCaseToSnakeCase	If TRUE, the data frame column names are assumed to use camelCase and are converted to snake_case before uploading.

Details

The databaseSchema argument is interpreted differently according to the different platforms: SQL Server and PDW: The databaseSchema schema should specify both the database and the schema, e.g. 'my_database.dbo'. Impala: the databaseSchema should specify the database. Oracle: The databaseSchema should specify the Oracle 'user'. All other : The databaseSchema should specify the schema.

This function sends the data in a data frame to a table on the server. Either a new table is created, or the data is appended to an existing table. NA values are inserted as null values in the database.

Bulk uploading:

Redshift: The MPP bulk loading relies upon the CloudyR S3 library to test a connection to an S3 bucket using AWS S3 credentials. Credentials are configured directly into the System Environment using the following keys: Sys.setenv("AWS_ACCESS_KEY_ID" = "some_access_key_id", "AWS_SECRET_ACCESS_KEY" = "some_secret_access_key", "AWS_DEFAULT_REGION" = "some_aws_region", "AWS_BUCKET_NAME" = "some_bucket_name", "AWS_OBJECT_KEY" = "some_object_key", "AWS_SSE_TYPE" = "server_side_encryption_type").

Spark (DataBricks): The MPP bulk loading relies upon the AzureStor library to test a connection to an Azure ADLS Gen2 storage container using Azure credentials. Credentials are configured directly into the System Environment using the following keys: Sys.setenv("AZR_STORAGE_ACCOUNT" = "some_azure_storage_account", "AZR_ACCOUNT_KEY" = "some_secret_account_key", "AZR_CONTAINER_NAME" = "some_container_name"). Prerequisites for Azure Databricks instances: Create an Access Connector for Azure Databricks to provide a secure bridge between Unity Catalog and Azure Data Lake Storage (ADLS Gen2), create the required storage credentials using the access connector, and configure the bulk-loading storage account as an external location using the access connector and storage credentials.

PDW: The MPP bulk loading relies upon the client having a Windows OS and the DWLoader exe installed, and the following permissions granted: –Grant BULK Load permissions - needed at a server level USE master; GRANT ADMINISTER BULK OPERATIONS TO user; –Grant Staging database permissions - we will use the user db. USE scratch; EXEC sp_addrolemember 'db_ddladmin', user; Set the R environment variable DWLOADER_PATH to the location of the binary.

PostgreSQL: Uses the 'psql' executable to upload. Set the POSTGRES_PATH environment variable to the Postgres binary path, e.g. 'C:/Program Files/PostgreSQL/11/bin' on Windows or '/Library/PostgreSQL/16/bin' on MacOS.

Examples

```
## Not run:
connectionDetails <- createConnectionDetails(
  dbms = "mysql",
  server = "localhost",
  user = "root",
```

```

    password = "blah"
  )
  conn <- connect(connectionDetails)
  data <- data.frame(x = c(1, 2, 3), y = c("a", "b", "c"))
  insertTable(conn, "my_schema", "my_table", data)
  disconnect(conn)

## bulk data insert with Redshift or PDW
connectionDetails <- createConnectionDetails(
  dbms = "redshift",
  server = "localhost",
  user = "root",
  password = "blah",
  schema = "cdm_v5"
)
conn <- connect(connectionDetails)
data <- data.frame(x = c(1, 2, 3), y = c("a", "b", "c"))
insertTable(
  connection = connection,
  databaseSchema = "scratch",
  tableName = "somedata",
  data = data,
  dropTableIfExists = TRUE,
  createTable = TRUE,
  tempTable = FALSE,
  bulkLoad = TRUE
) # or, Sys.setenv("DATABASE_CONNECTOR_BULK_UPLOAD" = TRUE)

## End(Not run)

```

isSqlReservedWord

Test a character vector of SQL names for SQL reserved words

Description

This function checks a character vector against a predefined list of reserved SQL words.

Usage

```
isSqlReservedWord(sqlNames, warn = FALSE)
```

Arguments

sqlNames	A character vector containing table or field names to check.
warn	(logical) Should a warn be thrown if invalid SQL names are found?

Value

A logical vector with length equal to sqlNames that is TRUE for each name that is reserved and FALSE otherwise

jdbcDrivers	<i>How to download and use JDBC drivers for the various data platforms.</i>
-------------	---

Description

Below are instructions for downloading JDBC drivers for the various data platforms. Once downloaded use the `pathToDriver` argument in the `connect()` or `createConnectionDetails()` functions to point to the driver. Alternatively, you can set the 'DATABASECONNECTOR_JAR_FOLDER' environmental variable, for example in your .Renv file (recommended).

SQL Server, Oracle, PostgreSQL, PDW, Snowflake, Spark, RedShift, Azure Synapse, BigQuery, InterSystems IRIS

Use the `downloadJdbcDrivers()` function to download these drivers from the OHDSI GitHub pages.

Netezza

Read the instructions [here](#) on how to obtain the Netezza JDBC driver.

Impala

Go to [Cloudera's site](#), pick your OS version, and click "GET IT NOW!". Register, and you should be able to download the driver.

SQLite

For SQLite we actually don't use a JDBC driver. Instead, we use the RSQLite package, which can be installed using `install.packages("RSQLite")`.

querySql	<i>Retrieve data to a data.frame</i>
----------	--------------------------------------

Description

This function sends SQL to the server, and returns the results.

Usage

```
querySql(
  connection,
  sql,
  errorReportFile = file.path(getOption("errorReportFileLocation", getwd()),
    "errorReportSql.txt"),
  snakeCaseToCamelCase = FALSE,
  integerAsNumeric = getOption("databaseConnectorIntegerAsNumeric", default = TRUE),
  integer64AsNumeric = getOption("databaseConnectorInteger64AsNumeric", default = TRUE)
)
```

Arguments

<code>connection</code>	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
<code>sql</code>	The SQL to be send.
<code>errorReportFile</code>	The file where an error report will be written if an error occurs. Defaults to 'errorReportSql.txt' in the directory given by <code>getOption("errorReportFileLocation")</code> , or the current working directory if that option is unset.
<code>snakeCaseToCamelCase</code>	If true, field names are assumed to use <code>snake_case</code> , and are converted to camel-Case.
<code>integerAsNumeric</code>	Logical: should 32-bit integers be converted to numeric (double) values? If FALSE 32-bit integers will be represented using R's native Integer class.
<code>integer64AsNumeric</code>	Logical: should 64-bit integers be converted to numeric (double) values? If FALSE 64-bit integers will be represented using <code>bit64::integer64</code> .

Details

Fields will be automatically converted for improved consistency in these situations:

- SQLite: Fields with names ending in `_date` will be converted to DATE fields. Rationale: SQLite does not support DATE fields.
- SQLite: Fields with names ending in `_datetime` will be converted to POSIXct fields. Rationale: SQLite does not support DATETIME fields.
- BigQuery and Snowflake: Integer fields will be converted to Integer if it fits in an integer, or will remain Integer64 otherwise. Rationale: these platforms do not distinguish between INT and BIGINT.

This function sends the SQL to the server and retrieves the results. If an error occurs during SQL execution, this error is written to a file to facilitate debugging. Null values in the database are converted to NA values in R.

Value

A data frame.

Examples

```
## Not run:
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = "localhost",
  user = "root",
  password = "blah",
  schema = "cdm_v4"
)
conn <- connect(connectionDetails)
count <- querySql(conn, "SELECT COUNT(*) FROM person")
disconnect(conn)

## End(Not run)
```

querySqlToAndromeda	<i>Retrieves data to a local Andromeda object</i>
---------------------	---

Description

This function sends SQL to the server, and returns the results in a local Andromeda object

Usage

```
querySqlToAndromeda(
  connection,
  sql,
  andromeda,
  andromedaTableName,
  errorReportFile = file.path(getOption("errorReportFileLocation", getwd()),
    "errorReportSql.txt"),
  snakeCaseToCamelCase = FALSE,
  appendToTable = FALSE,
  integerAsNumeric = getOption("databaseConnectorIntegerAsNumeric", default = TRUE),
  integer64AsNumeric = getOption("databaseConnectorInteger64AsNumeric", default = TRUE)
)
```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
sql	The SQL to be sent.
andromeda	An open Andromeda object, for example as created using <code>Andromeda::andromeda()</code> .
andromedaTableName	The name of the table in the local Andromeda object where the results of the query will be stored.
errorReportFile	The file where an error report will be written if an error occurs. Defaults to 'errorReportSql.txt' in the directory given by <code>getOption("errorReportFileLocation")</code> , or the current working directory if that option is unset.
snakeCaseToCamelCase	If true, field names are assumed to use snake_case, and are converted to camel-Case.
appendToTable	If FALSE, any existing table in the Andromeda with the same name will be replaced with the new data. If TRUE, data will be appended to an existing table, assuming it has the exact same structure.
integerAsNumeric	Logical: should 32-bit integers be converted to numeric (double) values? If FALSE 32-bit integers will be represented using R's native Integer class.
integer64AsNumeric	Logical: should 64-bit integers be converted to numeric (double) values? If FALSE 64-bit integers will be represented using <code>bit64::integer64</code> .

Details

Retrieves data from the database server and stores it in a local Andromeda object. This allows very large data sets to be retrieved without running out of memory. If an error occurs during SQL execution, this error is written to a file to facilitate debugging. Null values in the database are converted to NA values in R. If a table with the same name already exists in the local Andromeda object it is replaced.

Value

Invisibly returns the andromeda. The Andromeda object will have a table added with the query results.

Examples

```
## Not run:
andromeda <- Andromeda::andromeda()
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = "localhost",
  user = "root",
  password = "blah",
  schema = "cdm_v4"
)
conn <- connect(connectionDetails)
querySqlToAndromeda(
  connection = conn,
  sql = "SELECT * FROM person;",
  andromeda = andromeda,
  andromedaTableName = "foo"
)
disconnect(conn)

andromeda$foo

## End(Not run)
```

renderTranslateExecuteSql

Render, translate, execute SQL code

Description

This function renders, translates, and executes SQL consisting of one or more statements.

Usage

```
renderTranslateExecuteSql(
  connection,
  sql,
  profile = FALSE,
  progressBar = TRUE,
  reportOverallTime = TRUE,
```

```

    errorReportFile = file.path(getOption("errorReportFileLocation", getwd()),
                                "errorReportSql.txt"),
    runAsBatch = FALSE,
    tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
    ...
)

```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
sql	The SQL to be executed.
profile	When true, each separate statement is written to file prior to sending to the server, and the time taken to execute a statement is displayed.
progressBar	When true, a progress bar is shown based on the statements in the SQL code.
reportOverallTime	When true, the function will display the overall time taken to execute all statements.
errorReportFile	The file where an error report will be written if an error occurs. Defaults to 'errorReportSql.txt' in the directory given by <code>getOption("errorReportFileLocation")</code> , or the current working directory if that option is unset.
runAsBatch	When true the SQL statements are sent to the server as a single batch, and executed there. This will be faster if you have many small SQL statements, but there will be no progress bar, and no per-statement error messages. If the database platform does not support batched updates the query is executed as ordinarily.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.
...	Parameters that will be used to render the SQL.

Details

This function calls the render and translate functions in the `SqlRender` package before calling `executeSql()`.

Examples

```

## Not run:
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = "localhost",
  user = "root",
  password = "blah",
  schema = "cdm_v4"
)
conn <- connect(connectionDetails)
renderTranslateExecuteSql(connection,
  sql = "SELECT * INTO #temp FROM @schema.person;",
  schema = "cdm_synpuf"
)
disconnect(conn)

```

```
## End(Not run)
```

```
renderTranslateQueryApplyBatched
```

Render, translate, and perform process to batches of data.

Description

This function renders, and translates SQL, sends it to the server, processes the data in batches with a call back function. Note that this function should perform a row-wise operation. This is designed to work with massive data that won't fit in to memory.

The batch sizes are determined by the java virtual machine and will depend on the data.

Usage

```
renderTranslateQueryApplyBatched(
  connection,
  sql,
  fun,
  args = list(),
  errorReportFile = file.path(getOption("errorReportFileLocation", getwd()),
    "errorReportSql.txt"),
  snakeCaseToCamelCase = FALSE,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  integerAsNumeric = getOption("databaseConnectorIntegerAsNumeric", default = TRUE),
  integer64AsNumeric = getOption("databaseConnectorInteger64AsNumeric", default = TRUE),
  ...
)
```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
sql	The SQL to be send.
fun	Function to apply to batch. Must take data.frame and integer position as parameters.
args	List of arguments to be passed to function call.
errorReportFile	The file where an error report will be written if an error occurs. Defaults to 'errorReportSql.txt' in the directory given by <code>getOption("errorReportFileLocation")</code> , or the current working directory if that option is unset.
snakeCaseToCamelCase	If true, field names are assumed to use snake_case, and are converted to camel-Case.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

```
integerAsNumeric
    Logical: should 32-bit integers be converted to numeric (double) values? If
    FALSE 32-bit integers will be represented using R's native Integer class.
integer64AsNumeric
    Logical: should 64-bit integers be converted to numeric (double) values? If
    FALSE 64-bit integers will be represented using bit64::integer64.
...
    Parameters that will be used to render the SQL.
```

Details

Fields will be automatically converted for improved consistency in these situations:

- SQLite: Fields with names ending in `_date` will be converted to DATE fields. Rationale: SQLite does not support DATE fields.
- SQLite: Fields with names ending in `_datetime` will be converted to POSIXct fields. Rationale: SQLite does not support DATETIME fields.
- BigQuery and Snowflake: Integer fields will be converted to Integer if it fits in an integer, or will remain Integer64 otherwise. Rationale: these platforms do not distinguish between INT and BIGINT.

This function calls the render and translate functions in the `SqlRender` package before calling `querySql()`.

Value

Invisibly returns a list of outputs from each call to the provided function.

Examples

```
## Not run:
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = "localhost",
  user = "root",
  password = "blah",
  schema = "cdm_v4"
)
connection <- connect(connectionDetails)

# First example: write data to a large CSV file:
filepath <- "myBigFile.csv"
writeBatchesToCsv <- function(data, position, ...) {
  write.csv(data, filepath, append = position != 1)
  return(NULL)
}
renderTranslateQueryApplyBatched(connection,
  "SELECT * FROM @schema.person;",
  schema = "cdm_synpuf",
  fun = writeBatchesToCsv
)

# Second example: write data to Andromeda
# (Alternative to querySqlToAndromeda if some local computation needs to be applied)
bigResults <- Andromeda::andromeda()
writeBatchesToAndromeda <- function(data, position, ...) {
```

```

data$p <- EmpiricalCalibration::computeTraditionalP(data$logRr, data$logSeRr)
if (position == 1) {
  bigResults$rrs <- data
} else {
  Andromeda::appendToTable(bigResults$rrs, data)
}
return(NULL)
}
sql <- "SELECT target_id, comparator_id, log_rr, log_se_rr FROM @schema.my_results;"
renderTranslateQueryApplyBatched(connection,
  sql,
  fun = writeBatchesToAndromeda,
  schema = "my_results",
  snakeCaseToCamelCase = TRUE
)

disconnect(connection)

## End(Not run)

```

renderTranslateQuerySql

Render, translate, and query to data.frame

Description

This function renders, and translates SQL, sends it to the server, and returns the results as a data.frame.

Usage

```

renderTranslateQuerySql(
  connection,
  sql,
  errorReportFile = file.path(getOption("errorReportFileLocation", getwd()),
    "errorReportSql.txt"),
  snakeCaseToCamelCase = FALSE,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  integerAsNumeric = getOption("databaseConnectorIntegerAsNumeric", default = TRUE),
  integer64AsNumeric = getOption("databaseConnectorInteger64AsNumeric", default = TRUE),
  ...
)

```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
sql	The SQL to be send.
errorReportFile	The file where an error report will be written if an error occurs. Defaults to 'errorReportSql.txt' in the directory given by <code>getOption("errorReportFileLocation")</code> , or the current working directory if that option is unset.

```

snakeCaseToCamelCase
    If true, field names are assumed to use snake_case, and are converted to camel-
    Case.
tempEmulationSchema
    Some database platforms like Oracle and Impala do not truly support temp ta-
    bles. To emulate temp tables, provide a schema with write privileges where
    temp tables can be created.
integerAsNumeric
    Logical: should 32-bit integers be converted to numeric (double) values? If
    FALSE 32-bit integers will be represented using R's native Integer class.
integer64AsNumeric
    Logical: should 64-bit integers be converted to numeric (double) values? If
    FALSE 64-bit integers will be represented using bit64::integer64.
...
    Parameters that will be used to render the SQL.

```

Details

Fields will be automatically converted for improved consistency in these situations:

- SQLite: Fields with names ending in `_date` will be converted to DATE fields. Rationale: SQLite does not support DATE fields.
- SQLite: Fields with names ending in `_datetime` will be converted to POSIXct fields. Rationale: SQLite does not support DATETIME fields.
- BigQuery and Snowflake: Integer fields will be converted to Integer if it fits in an integer, or will remain Integer64 otherwise. Rationale: these platforms do not distinguish between INT and BIGINT.

This function calls the render and translate functions in the SqlRender package before calling `querySql()`.

Value

A data frame.

Examples

```

## Not run:
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = "localhost",
  user = "root",
  password = "blah",
  schema = "cdm_v4"
)
conn <- connect(connectionDetails)
persons <- renderTranslatequerySql(conn,
  sql = "SELECT TOP 10 * FROM @schema.person",
  schema = "cdm_synpuf"
)
disconnect(conn)

## End(Not run)

```

renderTranslateQuerySqlToAndromeda

Render, translate, and query to local Andromeda

Description

This function renders, and translates SQL, sends it to the server, and returns the results as an Andromeda object

Usage

```
renderTranslateQuerySqlToAndromeda(
  connection,
  sql,
  andromeda,
  andromedaTableName,
  errorReportFile = file.path(getOption("errorReportFileLocation", getwd()),
    "errorReportSql.txt"),
  snakeCaseToCamelCase = FALSE,
  appendToTable = FALSE,
  tempEmulationSchema = getOption("sqlRenderTempEmulationSchema"),
  integerAsNumeric = getOption("databaseConnectorIntegerAsNumeric", default = TRUE),
  integer64AsNumeric = getOption("databaseConnectorInteger64AsNumeric", default = TRUE),
  ...
)
```

Arguments

connection	The connection to the database server created using either <code>connect()</code> or <code>DBI::dbConnect()</code> .
sql	The SQL to be send.
andromeda	An open Andromeda object, for example as created using <code>Andromeda::andromeda()</code> .
andromedaTableName	The name of the table in the local Andromeda object where the results of the query will be stored.
errorReportFile	The file where an error report will be written if an error occurs. Defaults to 'errorReportSql.txt' in the directory given by <code>getOption("errorReportFileLocation")</code> , or the current working directory if that option is unset.
snakeCaseToCamelCase	If true, field names are assumed to use snake_case, and are converted to camel-Case.
appendToTable	If FALSE, any existing table in the Andromeda with the same name will be replaced with the new data. If TRUE, data will be appended to an existing table, assuming it has the exact same structure.
tempEmulationSchema	Some database platforms like Oracle and Impala do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

```

integerAsNumeric      Logical: should 32-bit integers be converted to numeric (double) values? If
                        FALSE 32-bit integers will be represented using R's native Integer class.
integer64AsNumeric    Logical: should 64-bit integers be converted to numeric (double) values? If
                        FALSE 64-bit integers will be represented using bit64::integer64.
...                   Parameters that will be used to render the SQL.

```

Details

This function calls the render and translate functions in the SqlRender package before calling [querySqlToAndromeda\(\)](#).

Value

Invisibly returns the andromeda. The Andromeda object will have a table added with the query results.

Examples

```

## Not run:
connectionDetails <- createConnectionDetails(
  dbms = "postgresql",
  server = "localhost",
  user = "root",
  password = "blah",
  schema = "cdm_v4"
)
conn <- connect(connectionDetails)
renderTranslatequerySqlToAndromeda(conn,
  sql = "SELECT * FROM @schema.person",
  schema = "cdm_synpuf",
  andromeda = andromeda,
  andromedaTableName = "foo"
)
disconnect(conn)

andromeda$foo

## End(Not run)

```

requiresTempEmulation *Does the DBMS require temp table emulation?*

Description

Does the DBMS require temp table emulation?

Usage

```
requiresTempEmulation(dbms)
```

Arguments

dbms	The type of DBMS running on the server. See connect() or createConnectionDetails() for valid values.
------	--

Value

TRUE if the DBMS requires temp table emulation, FALSE otherwise.

Examples

```
requiresTempEmulation("postgresql")
requiresTempEmulation("oracle")
```

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